

Zoidberg 2.0

Computer Aided Diagnosis

Pneumonia Detection from Chest X-Ray Images

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Item	Detail
Institution	Epitech Paris — MSc Information Technology
Goal	Binary + 3-class pneumonia classification from chest X-rays
Team	Jeet Ratanpara (Person A) + Tejaswini (Person B)
Dataset	5,856 chest X-ray images across 3 datasets
Jeet trained	Logistic Regression, Random Forest, MLP Neural Network
Tejaswini trained	SVM, KNN, CNN (binary) + SVM, KNN, CNN (3-class)
Best binary model	SVM (Tejaswini) — ROC-AUC 0.9952, 97.13% accuracy (Split 1)
Best of Jeet's models	MLP Neural Network — ROC-AUC 0.9937, best across all splits
Best 3-class model	CNN (Tejaswini) — 82.28% (Split 1), 72.28% (Split 2)

1. Project Overview

Zoidberg 2.0 is a machine learning project built for Epitech MSc IT. Given a chest X-ray image the system predicts whether the patient has pneumonia or not. We trained nine models in total — six for binary classification and three for the 3-class extension — compared them across three data splitting strategies, and built a Flask web application for live demonstration during the presentation.

The biggest technical challenge was class imbalance. Around 74% of images in the training set show pneumonia, so every model we first trained just predicted pneumonia for everything. Accuracy still came out at 74% on paper but normal recall was literally zero. Fixing this shaped most of our design decisions.

2. Who Did What

Jeet Ratanpara — Person A *(green rows throughout this document)*

Responsible for data setup, the full preprocessing pipeline, all three data splits, and three ML models: Logistic Regression, Random Forest, and MLP Neural Network. Also built the Flask web application.

Task	What was done
Project setup	Folder structure, venv, requirements.txt, GitHub init
Dataset organisation	Moved train/val/test into dataset1/2/3, checked NORMAL and PNEUMONIA subfolders
EDA	Counted images per dataset, plotted class distributions, showed sample X-rays
Preprocessing pipeline	Grayscale → resize 128×128 → normalise → flatten to 16,384 features
StandardScaler + PCA	Reduced 16,384 features to 100 components, retaining 87.88% of variance
Caching	Saved preprocessed arrays + fitted scaler/PCA to disk so nothing reruns
All 3 data splits	80/20 train–test, train/val/test with folders, 5-fold stratified CV
Logistic Regression	Trained on all 3 splits, full metrics + ROC curve + confusion matrix
Random Forest	Trained on all 3 splits, full metrics + ROC curve + confusion matrix
MLP Neural Network	Trained on all 3 splits, balanced version with image rotation augmentation

Image augmentation	Rotated Normal images at 10°/-10°/20°/-20° to fix imbalance (1,341 → 6,705)
Best model saved	MLP pipeline (scaler + PCA + classifier) saved as best_pipeline.joblib
Flask web application	Full local website for live demo: upload X-ray, pick model, see prediction

Tejaswini — Person B *(blue rows throughout this document)*

Responsible for SVM, KNN, and CNN for both binary and 3-class tasks, hyperparameter tuning, the final comparison table, and all project deliverables (notebook, HTML, PDF).

Task	What was done
SVM binary	RBF kernel, GridSearchCV with balanced_accuracy scoring, all 3 splits
KNN binary	K sweep 1–20, best K=9, all 3 splits with full metrics
CNN binary	3 Conv2D blocks, batch normalisation, dropout, threshold tuned to 0.7
SVM 3-class	One-vs-Rest strategy for Normal / Bacteria / Virus classification
KNN 3-class	Re-swept K for 3-class data, best K=18
CNN 3-class	Softmax output, sparse_categorical_crossentropy loss, class weights
Hyperparameter tuning	GridSearchCV on Dataset 2 (val) for all 4 sklearn models, optimised for ROC-AUC
Final comparison table	All models on official test set, sorted by ROC-AUC
Notebook integration	Merged all code into one clean runnable notebook
HTML + PDF export	jupyter nbconvert, synthesis PDF with goals/methods/results/conclusions
7 saved model files	All binary + 3-class models as .joblib and .keras

3. Dataset

Chest X-ray images from the Kaggle Chest X-Ray Pneumonia dataset, split into three separate folders. Each folder has NORMAL and PNEUMONIA subfolders.

Dataset	Folder	Role	Normal	Pneumonia	Total
Dataset 1	data/train/	Training (Jeet + Tejaswini)	1,341	3,875	5,216
Dataset 2	data/val/	Tuning (Tejaswini GridSearchCV)	8	8	16
Dataset 3	data/test/	Final evaluation only	234	390	624
TOTAL	—	—	1,583	4,273	5,856

4. Preprocessing Pipeline (Jeet)

Every image goes through 6 steps before reaching any model. This was all done by Jeet. CNN is the exception — it skips the PCA step and takes raw 64×64 pixels.

Step	Operation	Before	After	Why
1	Grayscale	RGB image	Black/white	X-rays carry no useful colour
2	Resize	Any size	128×128 px	All inputs must be same size
3	Normalise	Values 0–255	Values 0.0–1.0	ML models prefer small numbers
4	Flatten	128×128 grid	16,384 values	Sklearn models need 1D rows
5	StandardScaler	Varying ranges	Mean=0, Std=1	Stops large values dominating
6	PCA	16,384 features	100 components	Keeps 87.88% info, much faster training

Jeet also applied image rotation augmentation: Normal images were rotated at 10°, -10°, 20°, -20° to increase Normal training count from 1,341 to 6,705.

5. Jeet's Models (Logistic Regression, Random Forest, MLP)

Jeet trained three binary classification models, each evaluated on all three data splits. The same preprocessed data (scaler + PCA applied) was used for all three.

Logistic Regression

The simplest model. Finds a straight-line boundary between Normal and Pneumonia in the 100-dimensional PCA space. Used as the baseline.

Split	Accuracy	Precision	Recall	F1	ROC-AUC
Split 1 (80/20)	0.9598	0.9779	0.9678	0.9728	0.9891
Split 2 (official test)	0.7484	0.7161	0.9897	0.8310	0.8973
Split 3 (5-fold CV)	0.9559	—	—	0.9704	0.9871

Random Forest

An ensemble of 100 decision trees that vote on the final prediction. Expected to beat Logistic Regression but actually came in slightly lower on most splits, likely because PCA components are less interpretable as tree-split points.

Split	Accuracy	Precision	Recall	F1	ROC-AUC
Split 1 (80/20)	0.9416	0.9376	0.9871	0.9617	0.9858
Split 2 (official test)	0.7436	0.7114	0.9923	0.8287	0.9160
Split 3 (5-fold CV)	0.9329	—	—	0.9562	0.9833

MLP Neural Network ★ Best of Jeet's models

Two hidden layers of 256 and 128 neurons. MLP wins on every metric across all three splits compared to Logistic Regression and Random Forest. It learns complex non-linear patterns that a straight-line model cannot capture.

Split	Accuracy	Precision	Recall	F1	ROC-AUC
Split 1 (80/20)	0.9617	0.9779	0.9704	0.9741	0.9937 ★
Split 2 (official test)	0.7837	0.7466	0.9897	0.8512	0.9032
Split 3 (5-fold CV)	0.9680	—	—	0.9785	0.9928 ★

Jeet's three models ranked

Rank	Model	Split 1 AUC	Split 2 AUC	Split 3 AUC	Verdict
1st	MLP Neural Network	0.9937	0.9032	0.9928	Wins everywhere
2nd	Logistic Regression	0.9891	0.8973	0.9871	Close second, simpler
3rd	Random Forest	0.9858	0.9160	0.9833	Lowest, PCA hurts trees

6. Tejaswini's Models (SVM, KNN, CNN)

Tejaswini trained three binary models and three 3-class models. All binary models were tuned using GridSearchCV on Dataset 2 (val set).

SVM ★ Best binary model overall

Support Vector Machine with RBF kernel. The critical fix was changing GridSearchCV scoring from f1 to balanced_accuracy — this stopped the model from ignoring Normal images entirely.

Split	Accuracy	F1	ROC-AUC	Normal recall
Split 1 (80/20)	0.9713	0.9805	0.9952	97.01%
Split 2 (official test)	0.7949	0.8562	0.9161	49.15%
Split 3 (5-fold CV)	0.9682	0.9784	0.9952	~96%

KNN

K-Nearest Neighbours with K=9 (selected after sweeping K from 1 to 20). Simpler than SVM but surprisingly competitive. Decent Normal recall compared to other models.

Split	Accuracy	F1	ROC-AUC	Normal recall
Split 1 (80/20)	0.9617	0.9745	0.9809	89.93%
Split 2 (official test)	0.7676	0.8419	0.8741	~60%
Split 3 (5-fold CV)	0.9515	0.9677	0.9838	~88%

CNN ★ Best on real test data

Convolutional Neural Network with 3 convolutional blocks. Unlike all other models, CNN works on raw 64×64 pixels instead of PCA components — it learns actual visual features like lung texture and infiltrate shapes.

Configuration	Accuracy	F1	ROC-AUC	Normal recall	Pneumonia recall
Threshold 0.5	0.8542	0.8936	0.9499	64.53%	97.95%
Threshold 0.7 (chosen)	0.8574	0.8947	0.9499	67.09%	96.92%

CNN was only evaluated on Split 2. Running cross-validation on a deep learning model takes significantly more compute time, so it was tested on the official test set only.

7. Full Binary Model Comparison

All six models evaluated together.

Split 1 — Simple train/test split

Model	Person	Accuracy	F1	ROC-AUC
SVM (RBF)	Tejaswini	0.9713	0.9805	0.9952
MLP	Jeet	0.9617	0.9741	0.9937
KNN	Tejaswini	0.9617	0.9745	0.9809
LR	Jeet	0.9598	0.9728	0.9891
Random Forest	Jeet	0.9416	0.9617	0.9858

Split 2 — Official test set (most important)

Model	Person	Accuracy	F1	ROC-AUC
CNN	Tejaswini	0.8574	0.8947	0.9499
SVM	Tejaswini	0.7949	0.8562	0.9161
MLP	Jeet	0.7837	0.8512	0.9032
KNN	Tejaswini	0.7676	0.8419	0.8741
LR	Jeet	0.7484	0.8310	0.8973
Random Forest	Jeet	0.7436	0.8287	0.9160

Split 3 — 5-fold cross-validation

Model	Person	Accuracy	F1	ROC-AUC
SVM	Tejaswini	0.9682	0.9784	0.9952
MLP	Jeet	0.9680	0.9785	0.9928
LR	Jeet	0.9559	0.9704	0.9871
KNN	Tejaswini	0.9515	0.9677	0.9838
Random Forest	Jeet	0.9329	0.9562	0.9833

8. 3-Class Classification — Tejaswini

Tejaswini extended the project to classify Normal, Bacterial Pneumonia, and Viral Pneumonia. Three models were trained: SVM, KNN, and CNN. Jeet contributed the CNN Split 1 evaluation after the initial submission.

Model	Person	Split 1 Acc	Split 2 Acc	Split 3 CV
CNN	Tejaswini	0.8228	0.7228	N/A
SVM (OvR)	Tejaswini	0.8151	0.6939	0.7991
KNN (K=18)	Tejaswini	0.7835	0.7019	0.7697

CNN wins Split 1 and Split 2 making it the best 3-class model. SVM wins the cross-validation split. CNN N/A on Split 3 because running CV on a deep learning model takes too long on a laptop.

9. What Went Wrong and How We Fixed It

Problem	Who hit it	Fix
Normal recall stuck at 0%	Both	Jeet: rotation augmentation (1,341 → 6,705 Normal). Tejaswini: balanced_accuracy in GridSearchCV + class weights.
SVM predicted everything as Pneumonia	Tejaswini	Changed GridSearchCV scoring from f1 to balanced_accuracy. Normal accuracy went from 0% to 97%.
SVM + KNN too slow without PCA	Jeet	Applied PCA to reduce 16,384 features to 100 components. Training went from 30+ mins to seconds.
Validation set only 16 images	Tejaswini	Used Dataset 2 only as final held-out test, not for tuning. Actual parameter decisions used Split 3 CV.
TensorFlow not compatible with Python 3.14	Tejaswini	Installed TensorFlow on system Python 3.11 via Homebrew instead of the project venv.

10. Conclusions

Overall binary winners

- SVM (Tejaswini) — best overall with ROC-AUC 0.9952 and 97.13% accuracy. Wins 2 of 3 splits.
- CNN (Tejaswini) — best on real unseen test data at 85.74%, because it sees actual pixel patterns.

- MLP (Jeet) — best of Jeet's three models. Beats Logistic Regression and Random Forest on every single metric across all 3 splits.

Jeet model ranking

- 1st: MLP Neural Network — wins everything vs LR and RF
- 2nd: Logistic Regression — simple but very close
- 3rd: Random Forest — PCA components are not ideal for decision trees

3-class winner

- CNN (Tejaswini) wins 2 of 3 splits. SVM wins cross-validation. Virus was the hardest class in every model.

Lessons

- Evaluation metric in GridSearchCV matters. `balanced_accuracy` was the fix for class imbalance.
- PCA was essential. Not optional. Without it SVM and KNN were unusable.
- Split 2 is the honest number. 97% on Split 1 dropping to 79–85% on Split 2 is normal and expected.
- Cross-validation is more reliable than a single train/test split.